High-Efficiency Li-Ion Charger with Integrated Power Path Management and Protection

1 Features

- USB Type-C power input
- Single-cell 18650 Li-Ion battery support
- Integrated battery charger
- USB power output
- Battery level LED indicators
- Push-button control input
- Battery overcharge protection
- Battery overdischarge protection
- Resettable battery fuse
- Compact PCB implementation
- SMD component design

2 Applications

- Portable power banks
- Embedded electronic systems
- IoT devices
- Battery-powered equipment
- USB power delivery systems
- Portable instrumentation
- Development and prototyping platforms
- Low-power industrial electronics

3 Description

The Li-ion Battery Charger & Power Bank Module is a compact energy management solution designed for single-cell lithium-ion batteries. The system integrates battery charging, protection, and regulated 5V output generation into a single board.

The design is centered around a highly integrated power management IC, enabling simplified implementation while maintaining reliable operation. The module supports charging via a USB Type-C connector and provides a regulated 5V output suitable for powering external devices such as smartphones and embedded systems.

Special attention was given to power integrity and protection. The design includes local decoupling capacitors, bulk capacitance on the battery node, and a dedicated protection IC to prevent unsafe operating conditions.

Input protection is enhanced with an ESD diode, while a resettable fuse (PTC) provides additional safety against overcurrent conditions. The power path is optimized to minimize voltage drops and ensure stable operation during load transients.

The user interface consists of LED indicators displaying battery charge level and a tactile button for system activation and status indication.

This module is intended for prototyping, embedded applications, and as a reference design for battery-powered systems.

Block Diagram

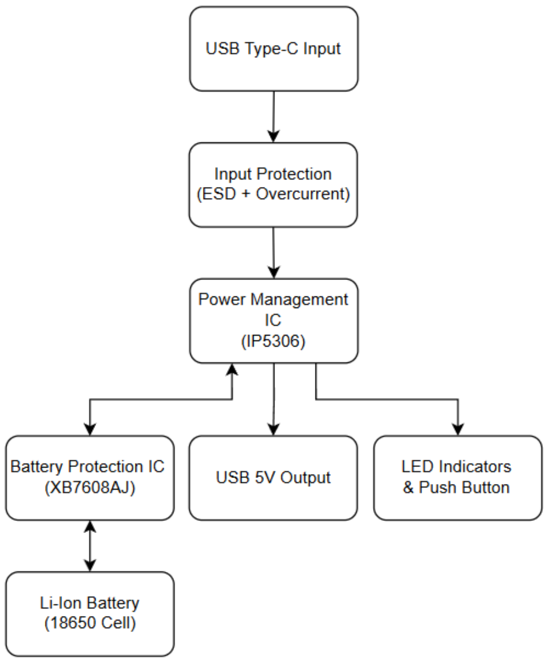

Figure 1. Functional Block Diagram

4 Device Information

Parameter	Value
Product Type	Power Management Module
Battery Configuration	Single-Cell Li-Ion
Nominal Battery Voltage	3.7 V
Battery Charge Voltage	4.2 V
Input Supply Voltage	5 V
Output Voltage	5 V
Input Connector	USB Type-C
Output Connector	USB Type-A
PMIC	IP5306
Protection IC	XB7608AJ
PCB Technology	2-Layer FR4
User Indicators	LED Battery Level Indicators
Protection Functions	OVP / UVP / OCP / SCP

5 PCB Stackup

Layer	Description
Top Layer	Power distribution, high-current routing, and component placement
Bottom Layer	Ground return path

6 PCB Layout

Top Layer

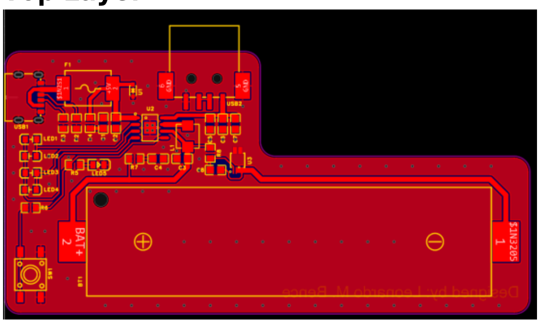

Figure 2. Top Layer

Bottom Layer

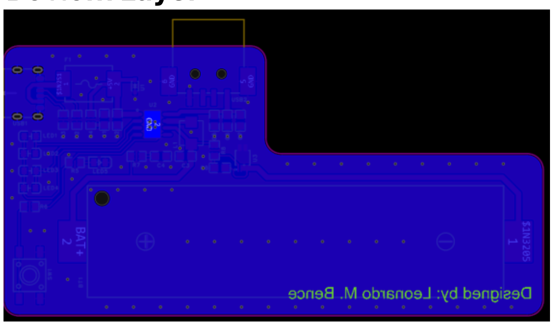

Figure 3. Bottom Layer

Note: The PCB layout was optimized for low impedance power distribution and stable converter operation. High-current paths were implemented using widened copper traces to reduce voltage drop and improve thermal performance during charging and boost converter operation.

7 Schematic

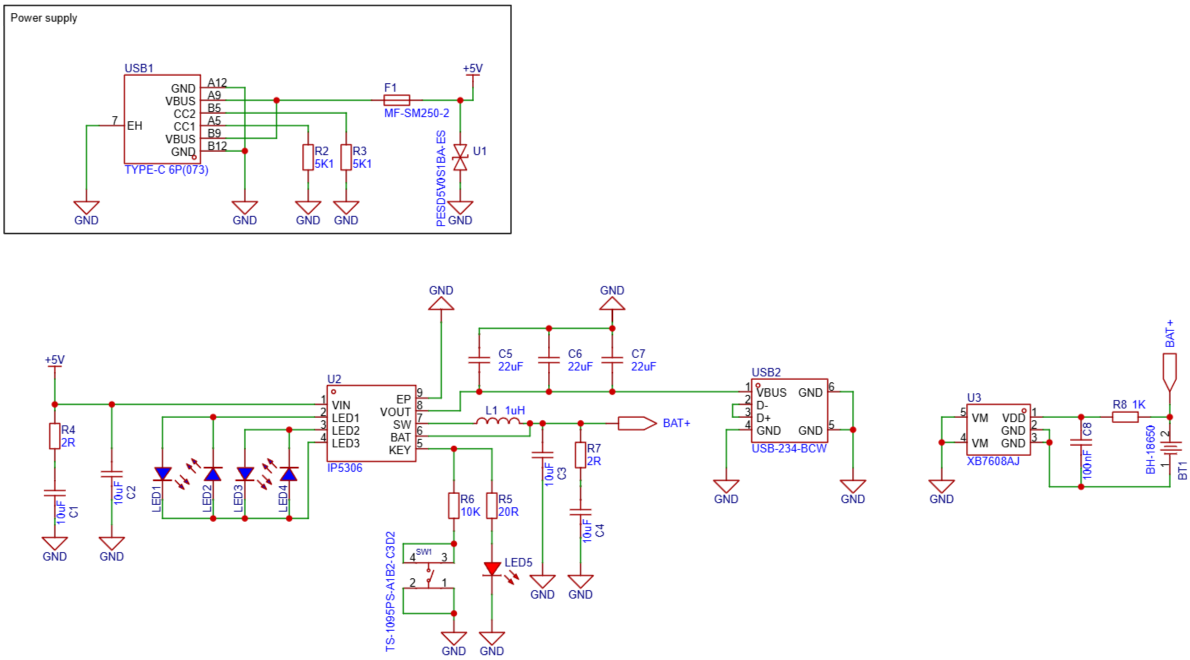

Figure 4. Schematic

8 BOM (Bill of Materials)

Name	Designator	Quantity
BH-18650	BT1	1
MF-SM250-2	F1	1
TS-1095PS-A1B2-C3D2	SW1	1
PESD5V0S1BA-ES	U1	1
XB7608AJ	U3	1
USB-234-BCW	USB2	1
100nF 50V 0805 X7R	C8	1
1K 0805	R8	1
1uH	L1	1
2R 0805	R4, R7	2
IP5306	U2	1
22uF 50V 0805 X7R	C5, C6, C7	3
5K1 0805	R2, R3	2
TYPE-C 6P(073)	USB1	1
LED 0805 BLUE	LED1, LED2, LED3, LED4	4
LED 0805 WHITE	LED5	1
20R 0805	R5	1
10K 0805	R6	1
10uF 50V 0805 X7R	C1, C2, C3, C4	4

9 Electrical Characteristics

Parameter	Conditions	Min	Typ	Max	Unit
Input Voltage	USB Type-C input	4.5	5.0	5.5	V
Battery Voltage	Single-cell Li-Ion	3.0	3.7	4.2	V
Battery Charge Voltage	Charging mode	—	4.2	—	V
USB Output Voltage	Boost converter enabled	4.8	5.0	5.2	V
Charging Current	Typical operation	—	2.0	—	A
Output Current	USB output	—	2.0	—	A
Standby Current	No load condition	—	—	100	µA
Boost Converter Efficiency	Typical load	—	90	—	%
Operating Temperature	Ambient	0	25	70	°C
Storage Temperature	—	-40	—	85	°C

10 Protection Features

Feature	Description
OVP	Battery overvoltage protection
UVP	Battery undervoltage protection
OCP	Charge and discharge overcurrent protection
SCP	Output short-circuit protection
OTP	Internal thermal shutdown protection
ESD Protection	USB input ESD suppression
PTC Fuse Protection	Resettable overcurrent battery protection

3D image of the PCB

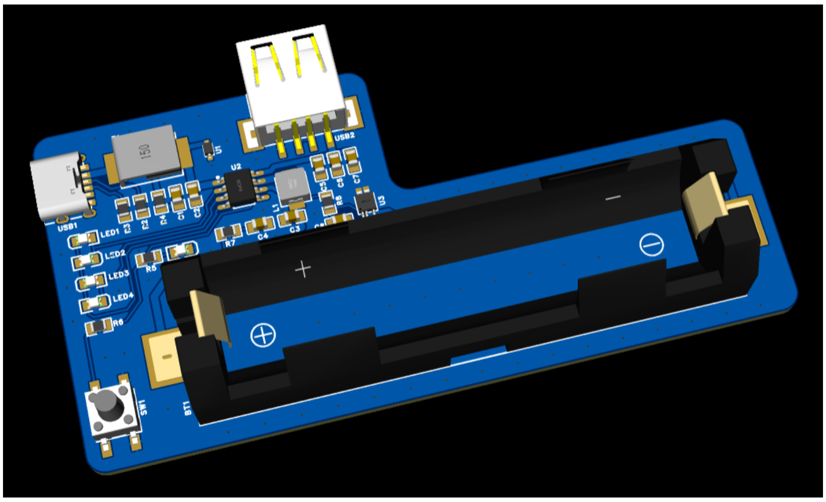

Figure 5. 3D PCB Image

PCB Assembled

Figure 6. PCB Assembled

Disclaimer

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